

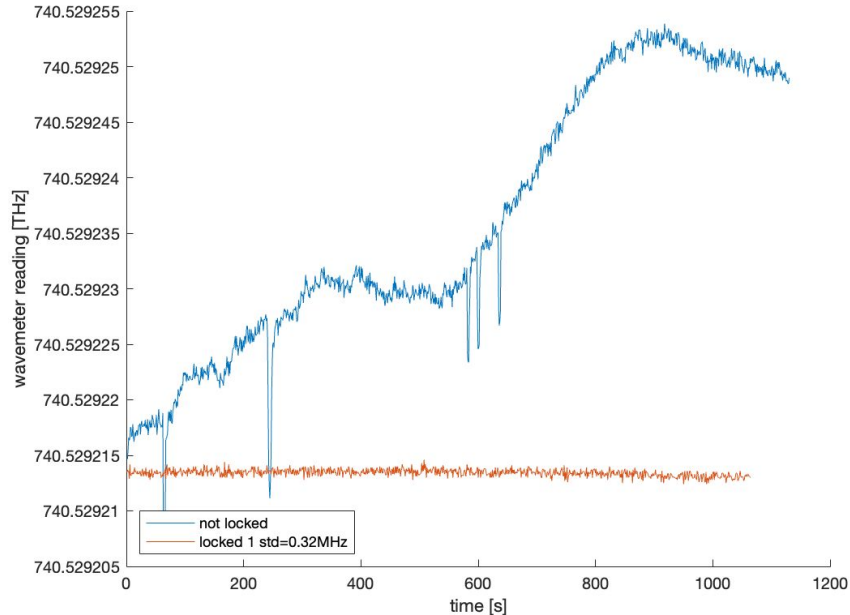
Locking state prep./EIT lasers

10/17/2022

Yuqi

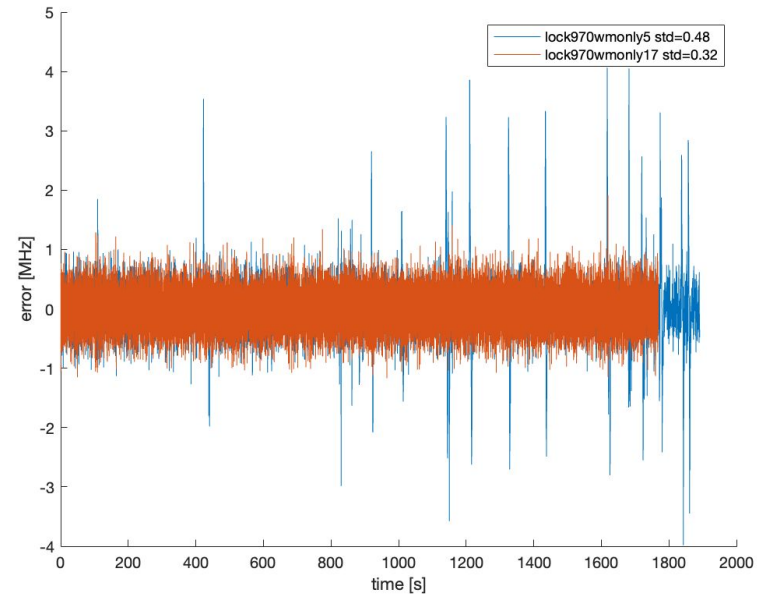
405 laser locked to transition

- doppler free peak, 20 kHz modulations
- piezo feedback



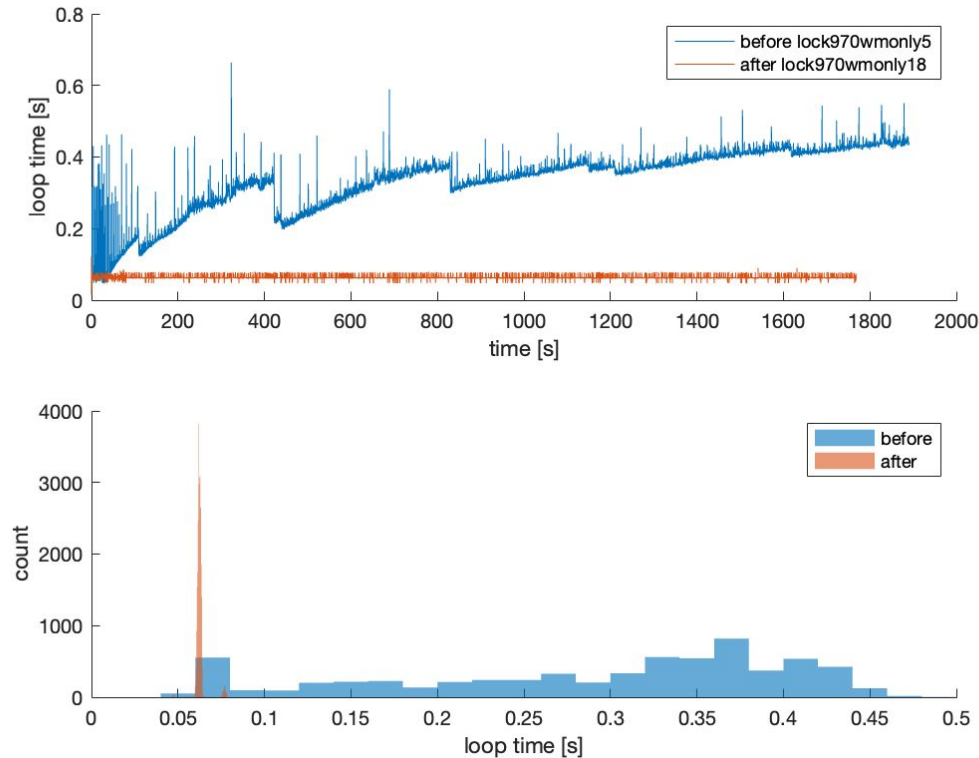
970 laser locked to wavemeter

- exposure time = 1 ms (= min exp.)
- software timed: loop time ~ 60 ms
- piezo feedback

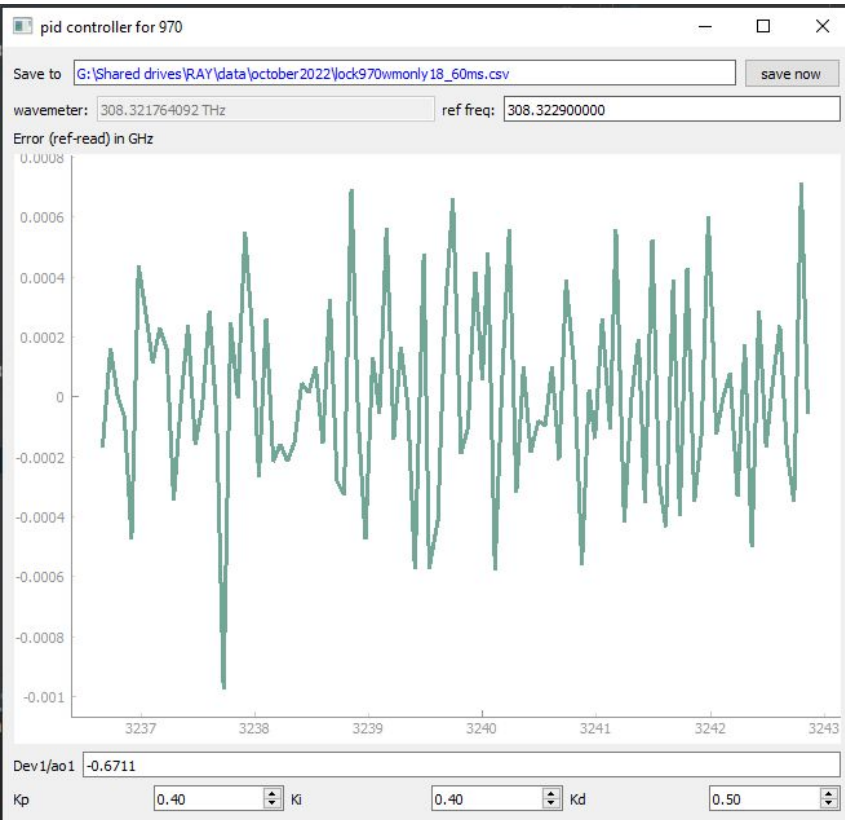


software loop time is somewhat critical for the software lock.

in each loop, wavemeter is read and pid correction is calculated and sent to USBdaq ao.
after changing a few things, the loop time became somewhat constant



pid controller for 970



- real time reading from wavemeter
- feedback voltage out: usbdaq ao1; sent to piezo via Fine 1 on dlc controller

to use this

- start reading on wavemeter's GUI
- run logger_laserlockv3.py to start locking automatically
- click 'save now' to save all data since start to a file anytime before closing
 - blue: new filename; gray: filename exists
 - time, error, and Vout will be saved in a table (csv w/ header)

other comments

- plot shows the last 100 data points ~ 6s
- everything except Dev1/ao1 can be changed in real time
 - Dev1/ao1 displays real time reading of pid out
- can update scan offset on dlc pro upon closing (pop up message)